



INTRODUCTION TO LONG WAVE EQUATIONS

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Long Wave Equations

- The waves considered in this section have relatively long periods and thus tend to have low d/L values so that they usually are shallow water waves even in the deeper ocean.
- As an alternative, we can start with the known physical characteristics of shallow water waves and directly develop the relevant form of the equations of continuity and motion.
- These equations are commonly known as the long wave equations.
- Owing to the typical applications of the long wave equations for the phenomena discussed in this chapter, these equations will be presented in three-dimensional cartesian form (x, y horizontal coordinates; z vertical coordinate) and the equations of motion will include the effects of surface and bottom stress as well as Coriolis acceleration.

- In the long wave equations, vertical flow velocities are assumed negligible.
- Horizontal flows are considered in terms of the average velocity over the vertical water column.
- This is done in terms of the volumetric flow rate per unit width of vertical section q_x and q_y for the two coordinate directions. Thus

$$q_x = \int_{-d}^{\eta} u dz$$

and

$$q_y = \int_{-d}^{\eta} v dz$$

- where d is the bottom depth below the still water level and η is the water surface elevation above the still water level.

- Considering a **mass balance** where the net sum of the horizontal flow rates into and out of a control volume equals the net change in volume per unit time of the control volume represented by a rise or fall of the water surface leads to the **equation of continuity** which can be written:

$$\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial \eta}{\partial t} = 0 \quad (1)$$

- To present the **equations of motion** for the two horizontal directions we must introduce the **bottom stresses in the x and y directions (τ_{bx} and τ_{by})** and the **surface stresses in the x and y directions (τ_{sx} and τ_{sy})**.
- These might typically represent bottom friction and the surface stress due to wind shear on the water. Also we must introduce the **Coriolis parameter f** where

$$f = 2\omega \sin \phi \quad (2)$$

- ω is the earth's rotational speed ($7.27 \times 10^{-5} \text{ rad/s}$)** and **Φ is the latitude** of the point on the earth's surface where the equations are being applied.

- This parameter would be zero at the equator and maximum at the poles.
- The product of f times a horizontal component of the flow velocity yields the horizontal Coriolis force per unit mass or acceleration which acts at right angles to that flow velocity component (to the right in the Northern hemisphere and to the left in the Southern hemisphere).
- The Coriolis acceleration is relatively small, but can be important in some long wave analyses where large masses of water are involved.
- Integrating the equations of motion from the bottom to the water surface yields a common form for the horizontal components of the equations of motion

$$-g(d + \eta) \frac{\partial \eta}{\partial x} + f q_y + \frac{1}{\rho} (\tau_{sx} - \tau_{bx}) = \frac{\partial}{\partial x} \left(\frac{q_x^2}{d + \eta} \right) + \frac{\partial}{\partial y} \left(\frac{q_x q_y}{d + \eta} \right) + \frac{\partial q_x}{\partial t} \quad (3a)$$

$$-g(d + \eta) \frac{\partial \eta}{\partial y} + f q_x + \frac{1}{\rho} (\tau_{sy} - \tau_{by}) = \frac{\partial}{\partial x} \left(\frac{q_x q_y}{d + \eta} \right) + \frac{\partial}{\partial y} \left(\frac{q_y^2}{d + \eta} \right) + \frac{\partial q_y}{\partial t} \quad (3b)$$

- where ρ is the density of the water. In Eqs. (3) the first term on the lefthand side represents the gravitational force per unit mass exerted through the slope of the water surface in that component direction; the second term is the Coriolis acceleration component; and the last term on the left represents the net horizontal stress per unit mass in that component direction.
- The three terms on the righthand side represent the resulting convective and local acceleration components for the water.
- As will be seen, in some physical situations certain terms in Eqs. (3) are negligible and may be excluded.
- Also, if the water surface fluctuations are small compared to the depth, the total water depth can be represented by d .
- The effects of a pressure gradient acting across the water surface are not included in the force summation on the lefthand side. If this force is significant in practical problems its effect is usually evaluated separately.

Tsunamis

- Tsunamis are a specific type of wave not caused by wind but by large, impulsive displacement of the sea level.
- The disturbance of the water surface is usually triggered by underwater earthquakes or underwater volcanic eruptions.
- The surface disturbance travels away from its origin in a pattern comparable with the patterns generated by the landing of a pebble in a pond.
- In deep water, tsunamis are not visible because they are small in height and very long in wavelength (periods ranging from 5 min to 60 min).
- They may grow to devastating proportions at the coast where the water is shallow.

1. One of the waves in a tsunami has a period of 20 min and a height of 0.6 m at a point in the ocean where the depth is 3800 m (this is the mean depth of the earth's seas). Determine the celerity and length of this wave. Determine its celerity, length, and height in a nearshore depth of 10 m assuming no refraction, diffraction, or reflection effects.

Solution:

Assume that the wave is a shallow water wave so

$$C = \sqrt{gd} = \sqrt{9.81(3800)} = 193 \text{ m/s (695 km/h)}$$

$$L = CT = 193(20)(60) = 231,700 \text{ m}$$

$$d/L = 3800/231,700 = 0.016$$

$$H/L = 2/231,700 = 8.6 \times 10^{-6}$$

Thus, the wave is a shallow water wave at a depth of 3800 m (and would be a shallow water wave in the deepest parts of the ocean). Note the large speed and length of the wave, and its very small steepness.

Equating wave power at the two water depths leads to

$$(H^2L)_{10} = (H^2L)_{3800}$$

Or, since

$$L = \sqrt{gd}T$$
$$(H^2\sqrt{d})_{10} = (H^2\sqrt{d})_{3800}$$

So, at a depth of 10 m

$$H = 0.6 \left(\frac{3800}{10} \right)^{0.25} = 2.65 \text{ m}$$

$$C = \sqrt{9.81(10)} = 9.9 \text{ m/s}$$

$$L = 9.9(20)60 = 11,900 \text{ m}$$

Shoaling has increased the wave height by a factor of $2.65/0.6 = 4.4$ times and refraction could cause a much greater increase.

Storm surges

- Storm surges are elevations of the water surface with time- and spatial scales equal to those of the large storm fields that generate them.
- They are generated by the low atmospheric pressure and high wind speeds in a storm field.
- The wavelength and period are generally slightly shorter than those of tides.
- Storm surges can cause severe flooding because the water will pile up against the coast when they approach the coast.
- Examples are the flooding of *New Orleans* due to the storm surge associated with hurricane *Katrina* in 2005, the regular cyclone-induced floodings of *Bangladesh* and the 1953 storm surge in the *Netherlands*.

- A storm over shallow nearshore coastal waters or shallow inland water bodies can generate large water level fluctuations if the storm is sufficiently strong and the water body is shallow over a large enough area.
- This is commonly known as storm surge or the meteorological tide.
- Storm activity can cause both a rise (setup) and fall (setdown) of the water level at different locations and times—with the setup commonly predominating in vertical magnitude, lateral extent, and duration.
- Specific causes of the water level change include: surface wind stress and the related bottom stress caused by currents generated by the surface wind stress, response to Coriolis acceleration as wind-induced currents develop, atmospheric horizontal pressure gradients, wind wave setup, long wave generation caused by the moving pressure disturbance, and precipitation and surface runoff.

- Storm surge calculations require a knowledge of the spatial and temporal distribution of surface wind speed and direction, and surface air pressure, as well as the forward path and speed of the design storm.
- Design wind and pressure field conditions for a site may be established by using the measured conditions from the worst storm of record in the general area.
- Or, if sufficient data exist, a return period analysis may be conducted to select storm design parameters (e.g., wind speed, pressure drop) having a specified frequency of occurrence for that area.
- Also, if sufficient historical surge elevation data are available, direct surge elevation–frequency relationships can be developed for a given area.
- A hurricane is a cyclonic storm having wind speeds in excess of 63 knots that originates in the tropics near but not at the equator.

- Those hurricanes that affect the United States typically move north from the region between Africa and South America (often with very irregular paths) into the Gulf of Mexico or up the Atlantic coast, eventually veering east and out over the Atlantic to dissipate.
- The driving mechanism is warm moist air that flows toward the eye (center) of the hurricane where the pressure is lowest.
- After rising in the eye the air flows outward at higher altitudes. Coriolis acceleration causes the inward flowing air to have a circular component, counterclockwise (looking down) in the northern hemisphere and clockwise in the southern hemisphere.
- The diameter of hurricanes usually does not exceed 300 nautical miles.
- The deficiency of warm moist air and the existence of increased surface friction when the hurricane is over land will cause it to dissipate, as will the deficiency of warm moist air at higher ocean latitudes.

- The surface pressure in a hurricane, which decreases from the ambient pressure at the periphery P_a to the lowest pressure at the eye P_e can be estimated by the following empirical equation (see Schwerdt et al., 1979)

$$\frac{p_r - p_e}{p_a - p_e} = e^{-R/r} \quad (4)$$

- which is based on an analysis of historical hurricane pressure data.
- In Eq. (4), P_r is the pressure at any radius r from the eye of the hurricane and R is the radius from the eye to the point of maximum wind speed.
- Wind speeds increase from the periphery toward the eye with maximum wind speeds occurring typically at a radius of between 10 and 30 nautical miles from the eye (where the pressure gradient is the largest).
- Inside this point the wind speeds diminish rapidly to near calm conditions at the eye. The pressure at the periphery may be taken as about the standard atmospheric pressure of 29.92 inches of mercury.

- The pressure in the eye varies significantly, naturally being lower for stronger hurricanes.
- A pressure in the eye of the order of 27.5 inches of mercury will occur in a strong hurricane and pressures as low as near 26 inches of mercury having been recorded.
- The hurricane season in the north Atlantic is from June to November with most hurricanes occurring in the months of August to October.

Standard Project Hurricane (*SPH*) and Probable Maximum Hurricane (*PMH*)

- Hypothetical design hurricanes for the U.S. Atlantic and Gulf coasts.
- The *SPH* has “a severe combination of values of meteorological parameters that will give high sustained wind speeds reasonably characteristic of a specified location.”
- Only a few hurricanes of record for a large region covering the area of concern will exceed the *SPH*.
- The combined return period for the total wind field will be several hundred years.
- The *PMH* has “a combination of values of meteorological parameters that will give the highest sustained wind speed that can probably occur at a specified coastal location.”
- The *PMH* was developed largely for use in the design of coastal nuclear power plants.
- Besides being used for storm surge analysis the *SPH* and *PMH* are useful for wave prediction and the analysis of wind loads on coastal structures.

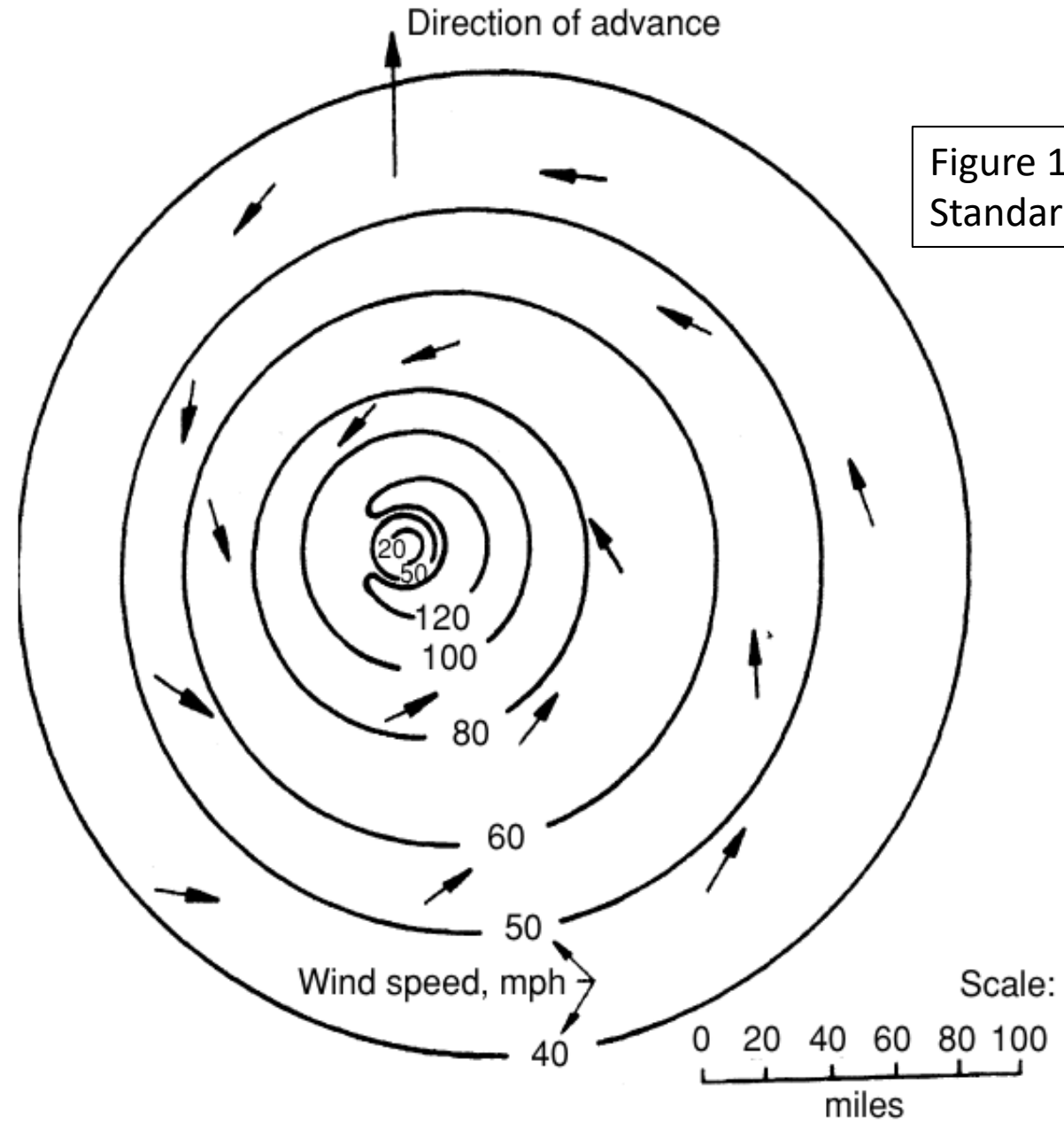


Figure 1: Typical surface wind field for a Standard Project Hurricane.

- The *SPH* and *PMH* are presented in terms of the central pressure in the eye, the peripheral pressure at the outer boundary, the radius to maximum wind speed, the forward speed of the hurricane, the direction of movement as the hurricane approaches the coast, and the inflow angle which is the angle between the wind direction and a circle concentric with the hurricane eye (The latter typically varies from 10 to 30 degrees at points from the location of maximum wind speed to the hurricane periphery).
- Ranges of values for the listed hurricane parameters are given as a function of coastal location from Maine to Texas.
- With selected values of these parameters the surface (10 m elevation) wind field can be plotted and the pressure field can be calculated from Eq. (4). Figure 1 shows the surface wind field for a typical SPH.

Numerical Analysis of Storm Surge

- A wind stress drag coefficient is required to convert the surface wind speed to a resulting surface stress for the equations of motion.
- A commonly used relationship was developed by Van Dorn (1953).
- From elementary fluid mechanics the wind-induced shear stress on a surface is typically written as a product of the drag coefficient times the air density times the wind speed squared.
- For storm surge analysis it is more useful to write a relationship for surface stress in terms of the water density ρ , i.e.,

$$\tau_s = K_s \rho W^2 \quad (5)$$

- where K_s is the wind stress drag coefficient and W is the wind speed at the standard 10 m elevation. The relationship given by Van Dorn is

$$K_s = 1.21 \times 10^{-6} + 2.25 \times 10^{-6} \left(1 - \frac{5.6}{W}\right)^2 \quad (6)$$

- where the wind speed is in meters per second.

Wind/Bottom Stress Setup

- The wind acting on the water surface causes a shear stress given by Eq. (5), where the wind stress drag coefficient K_s can be defined by Eq. (6) or some other relationship as discussed above.
- The surface wind stress generates a current that, in turn, develops a bottom stress.
- Usually, the bottom current velocity is not known nor can it easily be calculated, so the bottom stress cannot be directly calculated.
- Van Dorn (1953), from his field data and the threshold limits on his equipment for measuring the bottom stress, reported that it was generally less than 10% of the surface stress.
- Thus, the ratio $\tau_b/\tau_s = 0.1$ can be used for wind induced setup calculations if no better data are available.
- If adequate wind, pressure, and resulting setup data are available from a previous storm at a particular site, it is better to determine a local combined surface and bottom stress coefficient $K_{sb}(\tau_s + \tau_b = K_{sb}\rho W^2)$, which then acts as an "all inclusive" calibration factor for subsequent wind/bottom stress setup calculations.

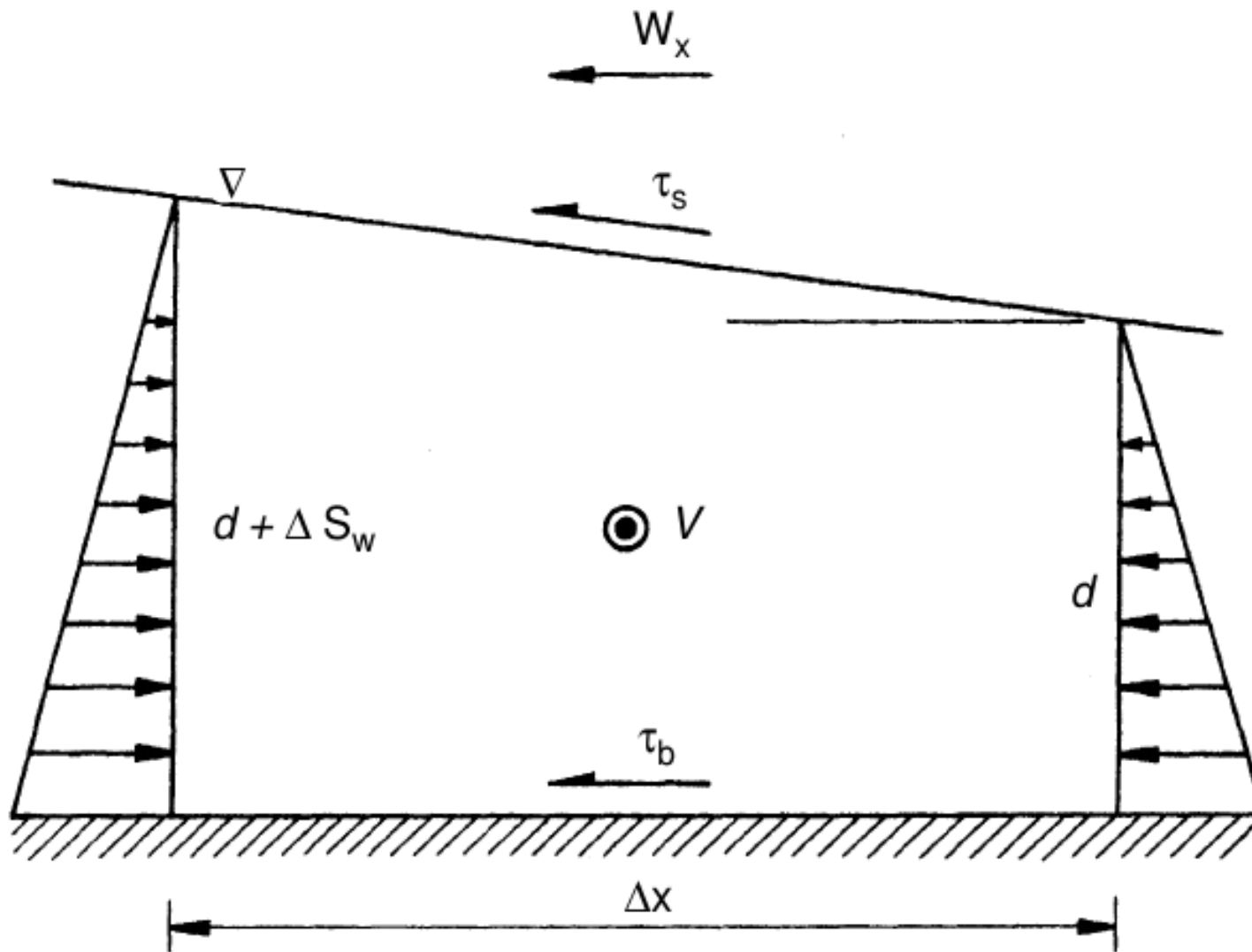


Figure 2: Definition sketch for wind/bottom stress and Coriolis setup derivations.

- Figure 2 shows a section of a nearshore water segment of length Δx in the direction in which setup is being calculated, unit width, and depths d and $d + \Delta S_w$.
- ΔS_w is the setup owing to wind and bottom stresses acting over the distance Δx .
- The hydrostatic forces on each end are shown and the bottom stress is drawn in the same direction as the surface wind stress because it is assumed that the wind-generated current causes a reverse flow on the bottom.
- If the wind is blowing at an angle θ to the x direction, then the component of the wind velocity in the x direction W_x is used.
- The effective stress then becomes

$$\tau_s + \tau_b = K_{sb} \rho W^2 \cos \theta = K_{sb} \rho W W_x$$

- Since the water surface slope $\Delta S_w / \Delta x$ is extremely flat a static balance yields

$$\tau_s \Delta x + \tau_b \Delta x + \frac{1}{2} g \rho d^2 - \frac{1}{2} g \rho (d + \Delta S_w)^2 = 0$$

- Inserting the wind/bottom stress relationship, eliminating higher order terms, and solving yields

$$\Delta S_w = d \left[\sqrt{\frac{2K_{sb}WW_x\Delta X}{gd^2} + 1} - 1 \right] \quad (7)$$

- where $K_{sb} = 1.1K_s$ may be used if no better information is available.

- For a shore normal profile on the open coast, wind/bottom stress setup calculations can be made using Eq. (7), keeping in mind the assumptions discussed above.
- The profile would be broken into segments (not necessarily of equal size) of length Δx over which an average depth and wind velocity can be defined.
- Starting at the offshore point where the wind velocity is sufficiently low and/or the water is sufficiently deep so setup is negligible, the setup for each segment can be calculated and summed to yield the total setup at the shore.
- As the calculation proceeds, the accumulated setup for the previous segments should be included in determining the water depth for the next segment setup calculation.

- Equation (7) can be applied (with caution) to enclosed bays that are not too irregular in shape.
- The depth for a given segment would have to be the average over the length and width of the segment.
- The volume of water that sets up on the leeward end of the bay must equal that removed from the windward end and the two sections would be divided by a nodal line.
- For a given wind field, a nodal line position would be assumed and the setup calculations would be carried out up wind and down wind from the nodal line.
- Then the setup and setdown water volumes would be determined and compared.
- A balance of the setup and setdown volumes determines the final surface profile and nodal line position.

Atmospheric Pressure Gradient Setup

- From hydrostatics, the water level variation or setup S_p owing to a horizontal atmospheric pressure differential Δp between two points on a continuous body of water is

$$S_p = \frac{\Delta p}{\rho g} \quad (8)$$

- where ρ is the water density.
- By combining Eqs. (4) and (8) one can calculate the static pressure setup that occurs at a point of interest in an SPH.
- For a pressure differential of say 2.5 inches of mercury (e.g., 29.9 to 27.4) from the periphery to the eye of a hurricane, the static pressure setup at the eye would be 0.86 m

2. The bottom profile normal to the shoreline in the area between Galveston and Port Arthur, TX is tabulated below (depths were read in fathoms and converted to meters; 1 fathom = 6 feet = 1.83 m).

Depth (m)	Distance (m)
7.31	2750
10.05	4590
13.72	22930
14.62	45860
20.12	48150
23.97	82550
25.60	100890
27.43	107710
36.57	121530
54.86	144460
73.15	181150
91.43	190320

Seaward of the 91.43 m depth the depth increases relatively rapidly and is sufficiently deep so that little additional setup would occur.

For a 60knot (30.86 m/s) wind blowing landward along this profile calculate the setup on the shore. If the pressure gradient along this profile is 0.003 inches of mercury per nautical mile what pressure setup would occur?

From Eq. (6) the drag coefficient is

$$K_s = 1.21 \times 10^{-6} + 2.25 \times 10^{-6} \left[1 - \frac{5.6}{30.86} \right]^2 = 2.72 \times 10^{-6}$$

Then, Eq. (7) becomes

$$\Delta S_w = d \left[\sqrt{\frac{2(1.1)(2.72)10^{-6}(30.86)^2 \Delta x}{9.81d^2} + 1} - 1 \right]$$

or

$$\Delta S_w = d \left[\sqrt{\frac{0.00058 \Delta x}{d^2} + 1} - 1 \right]$$

We then break the profile into segments having a length Δx and a depth $d + \Sigma \Delta S_w$ which is the depth below still water for the segment plus the accumulated surge up to that segment. The accumulated surge is then calculated from the above equation, starting at the seaward segment and using the depth plus accumulated surge for d . The results are tabulated below (all values are in meters):

d	Δx	ΔS_w	$d + \Sigma \Delta S_w$
3.66	2750	0.14	5.74
8.68	1840	0.04	10.72
11.88	18340	0.39	13.53
14.17	22930	0.43	15.39
17.37	2290	0.04	18.55
21.95	34400	0.44	22.69
24.79	18340	0.21	25.32
26.51	6820	0.07	26.97
32.00	13820	0.12	32.34
45.72	22930	0.14	45.92
64.00	36690	0.17	64.03
82.29	9170	0.03	82.29

Thus, the total wind-induced setup at the shore would be 2.22 m ($\Sigma \Delta S_w$) and the setup profile could be determined by sequentially summing the values in the second and third columns.

The distance from the offshore starting point to the shore is 190320 m ($\Sigma \Delta x$) = 102.8 nautical miles. Thus, the pressure difference is $(0.003)102.8 = 0.31$ inches of mercury. From hydrostatics

$$S_p = \frac{0.31(13.55)}{12(3.28)} = 0.11 \text{ m}$$

Note how the above calculation relates to Eq. (8).

Coriolis Setup

- If a current is flowing along a shoreline, Coriolis acceleration will cause the water to deflect to the right in the Northern hemisphere.
- If the shoreline is on the righthand side of the current the deflection will be restrained, causing a setup of water on the shore.
- If the shoreline is on the lefthand side there will be a setdown at the shore.
- In Figure 2 a current of velocity V is coming out of the page and Coriolis acceleration causes a setup as shown.
- A balance between the hydrostatic forces and the Coriolis acceleration or force per unit mass yields

$$\frac{1}{2} g \rho d^2 + 2 \omega V \sin \phi \rho g d \Delta x - \frac{1}{2} g \rho (d + \Delta S_c)^2 = 0 \quad (9)$$

- where S_c is the Coriolis set up. Neglecting higher order terms leads to

$$S_c = \frac{2\omega}{g} V \sin \phi \Delta x \quad (10)$$

- which can be used to calculate the setup over a given horizontal distance at any latitude if the current velocity is known.
- Away from the shore, wind-generated currents are free to respond to the Coriolis acceleration; but nearshore this response is restricted and setup or setdown occurs.
- For simple analyses the resulting wind-generated current velocity is not easily determined.
- Often, the current velocity is small or in a direction such that the Coriolis setup can be neglected in a simple analysis. The latter is often true in small bays and lakes.

- Bretschneider (1967) solved the equation of motion for the alongshore direction, including only the wind and bottom stresses and the local acceleration to obtain an equation for the wind-generated longshore current velocity.
- For the resulting steady-state condition (i.e., sufficient wind duration)

$$V = W d^{1/6} \sqrt{\frac{K_s}{14.6n^2} \sin \theta} \quad (11)$$

- where n is Manning's roughness coefficient (typical value is 0.035), θ is the angle between the wind direction and a line perpendicular to the coast and/or bottom contours, and V is the resulting alongshore current velocity.